

Casella / Berger - Problems

7.1 One observation is taken on a discrete random variable X with PMF $f(x|\theta)$, where $\theta \in 1, 2, 3$. Find the MLE of θ .

x	$f(x 1)$	$f(x 2)$	$f(x 3)$
0	$1/3$	$1/4$	0
1	$1/3$	$1/4$	0
2	0	$1/4$	$1/4$
3	$1/6$	$1/4$	$1/2$
4	$1/6$	0	$1/4$

With one observation $X=x \rightarrow L(\theta|x) = f(x|\theta)$, $\theta \in \{1, 2, 3\}$

(From the likelihood definition, specializing to $n=1$) so for each possible θ , the
Now let's compare likelihood values for each possible observed x

for each observation $x \in \{0, 1, 2, 3, 4\} \rightarrow L(1|x) = f(x|1)$, $L(2|x) = f(x|2)$, $L(3|x) = f(x|3)$
And then choose θ that gives the largest value.

For $x=0$. From the table $f(0|1) = \frac{1}{3}$, $f(0|2) = \frac{1}{4}$, $f(0|3) = 0$

$$L(1|0) = \frac{1}{3}, \quad L(2|0) = \frac{1}{4}, \quad L(3|0) = 0$$

The largest $\rightarrow \hat{\theta}(0) = 1$

$$x=1 \rightarrow f(1|1) = \frac{1}{3}, \quad f(1|2) = \frac{1}{4}, \quad f(1|3) = 0$$

$\hat{\theta}(1) = 1$

$$x=2 \rightarrow f(2|1) = 0, \quad f(2|2) = \frac{1}{4}, \quad f(2|3) = \frac{1}{4}$$

$$L(1|2) = 0, \quad L(2|2) = \frac{1}{4}, \quad L(3|2) = \frac{1}{4}$$

largest value is $\frac{1}{4}$, achieved by both $\theta=2$ and $\theta=3$

So the MLE is not unique here $\hat{\theta}(2) \in \{2, 3\}$

$$x=3 \rightarrow f(3|1) = \frac{1}{6}, \quad f(3|2) = \frac{1}{4}, \quad f(3|3) = \frac{1}{2}$$

$$L(1|3) = \frac{1}{6}, \quad L(2|3) = \frac{1}{4}, \quad L(3|3) = \frac{1}{2}$$

$$\hat{\theta}(3) = 3$$

$$x=4 \rightarrow f(4|1) = \frac{1}{6}, \quad f(4|2) = 0, \quad f(4|3) = \frac{1}{4}$$

$\hat{\theta}(4) = 3$

$$\Rightarrow \hat{\theta}(x) = \begin{cases} 1, & x=0 \text{ or } 1 \\ \{2, 3\} & x=2 \text{ (tie)} \\ 3, & x=3 \text{ or } 4. \end{cases}$$

7.11 Let $X_1, X_2, \dots, X_n$ be iid with PDF. $f(x|\theta) = \theta x^{\theta-1}$, $0 \leq x \leq 1$, $0 < \theta < \infty$

a) Find the MLE of θ , and show that its invariance $\rightarrow 0$ as $n \rightarrow \infty$

b) Find the method of moments estimator of θ

② MLE and show $\text{Var}(\hat{\theta}) \rightarrow 0$.

① Build the likelihood $L(\theta|x)$. Because the sample is iid,

$$L(\theta|x_1, \dots, x_n) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \theta x_i^{\theta-1}$$

Now simplify:

$$L(\theta|x) = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}$$

② - Take logs

$$\ell(\theta) = \log L(\theta|x) = \log \left(\theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1} \right) = n \log \theta + (\theta-1) \log \left(\prod_{i=1}^n x_i \right)$$

$$\text{Since } \log \left(\prod x_i \right) = \sum \log x_i \rightarrow$$

$$\ell(\theta) = n \log \theta + (\theta-1) \sum_{i=1}^n \log x_i.$$

③ - Differentiate, set equal to 0, solve

Differentiate:

$$\ell'(\theta) = \frac{n}{\theta} + \sum_{i=1}^n \log x_i$$

Set $\ell'(\theta) = 0$:

$$\frac{n}{\hat{\theta}} + \sum_{i=1}^n \log x_i = 0 \rightarrow \frac{n}{\hat{\theta}} = - \sum_{i=1}^n \log x_i \rightarrow \hat{\theta} = \left(-\frac{1}{n} \sum_{i=1}^n \log x_i \right)^{-1}$$

④ - Second derivative check (confirmation of Max) $\rightarrow$

$$\frac{n}{\theta} = n \theta^{-1} \rightarrow \text{power rule: } \frac{d}{d\theta} \theta^k = k \theta^{k-1} \rightarrow \frac{d}{d\theta} [n \theta^{-1}] = n \cdot (-1) \theta^{-2} = -n \theta^{-2} = -\frac{n}{\theta^2}$$

$$\frac{d}{d\theta} \left[\sum_{i=1}^n \log x_i \right] = 0 \rightarrow \text{combine } \ell''(\theta) = -\frac{n}{\theta^2} < 0, \text{ so } \ell \text{ is concave and this critical point is the (MLE)}$$

⑤ - To make the denominator into positive and interpretable

Because $0 \leq X_i \leq 1$, we have $\log X_i \leq 0$ so $-\log X_i \geq 0$, meaning $Y_i \geq 0$, $T = \sum_{i=1}^n Y_i \geq 0$

so $\hat{\theta} = \frac{n}{T} > 0$, matching the parameter restriction $\theta > 0$

Part (b) Find the Method of Moments estimator of θ . for $f(x|\theta) = \theta x^{\theta-1}$, $0 \leq x \leq 1$, $\theta > 0$
 $X \sim \text{Beta}(\theta, 1)$

Compute $E[X] = \frac{\theta}{\theta+1}$

Then it solves $\bar{X} = \frac{\theta}{(\theta+1)} \rightarrow \bar{X}(\theta+1) = \theta \Rightarrow \bar{X}\theta + \bar{X} = \theta \rightarrow \bar{X} = \theta(1-\bar{X}) \rightarrow \bar{\theta} = \frac{\bar{X}}{1-\bar{X}}$

In terms of the sum

$$\bar{\theta} = \frac{\frac{1}{n} \sum_{i=1}^n X_i}{1 - \frac{1}{n} \sum_{i=1}^n X_i} = \frac{\sum_{i=1}^n X_i}{n - \sum_{i=1}^n X_i}$$

Wasserman

Section 9.14

Problem 2. Let $X_1, X_2, \dots, X_n \sim \text{Uniform}(a, b)$, where a and b are unknown parameters and $a < b$.

(a) Find the method of moments estimator for a and b .

(b) Find the MLE estimates of a and b .

(c) Let $\tau = \int x dF(x)$. find the MLE of τ

(d) Let $\hat{\tau}$ be the MLE of τ . Let $\tilde{\tau}$ be the non-parametric plug in estimator of $\tau = \int x dF(x)$ suppose that $a=1$, $b=3$, and $n=10$. Find the MSE of $\hat{\tau}$ by simulation. Find the MSER of $\tilde{\tau}$ analytically and compare to the simulated value.

Part (a) To estimate two parameters (a, b) using MoM, we match a "center" summary (mean) and a "spread" summary (variance). Because $\text{Uniform}(a, b)$ is completely determined by its midpoint and width. Here $\theta = (a, b)$ has two unknowns, so we need two equations.

Wasserman defines the j -th population moment and j -th sample moment by

$\alpha_j(\theta) = E_{\theta}(X^j)$, $\hat{\alpha}_j = \frac{1}{n} \sum_{i=1}^n X_i^j$ and the MOME $\hat{\theta}_{\text{MOM}}$ is the value of θ that solves the sys.
 $\alpha_1(\hat{\theta}) = \hat{\alpha}_1$, $\alpha_2(\hat{\theta}) = \hat{\alpha}_2, \dots, \alpha_n(\hat{\theta}) = \hat{\alpha}_n$

Step (1) Population mean and variance for $\text{Uniform}(a, b)$

$E[X] = \frac{a+b}{2}$, $\text{Var}(X) = \frac{(b-a)^2}{12}$

Mean $E[X] = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}$

Variance 1) compute $E[X^2] = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)}$

use $b^3 - a^3 = (b-a)(a^2 + ab + b^2) \rightarrow E[X^2] = \frac{a^2 + ab + b^2}{3}$

So let's turn the problem into "sum of exponentials"

⑤ Re-express $\hat{\theta}$ using $Y_i = -\log X_i$

We define $Y_i = -\log X_i$ ($x \geq 0$) Then $-\sum_{i=1}^n \log X_i = \sum_{i=1}^n Y_i \equiv T$ so $\hat{\theta} = \frac{n}{T}$

⑥ Find the distribution of $T = \sum Y_i$

$T = \sum_{i=1}^n Y_i \sim \text{Gamma}(n, \frac{1}{\theta})$ so from now on: $T \sim \Gamma(n, \frac{1}{\theta})$
 $\hat{\theta} = \frac{n}{T}$

⑦ Compute $E(\frac{1}{T})$ and $E(\frac{1}{T^2})$

7.1 Compute $E[\frac{1}{T}]$

For $T \sim \Gamma(n, \text{rate } \theta)$, the PDF is $f_T(t) = \frac{\theta^n}{\Gamma(n)} t^{n-1} e^{-\theta t}$, $t > 0$

$E[g(T)] = \int_0^\infty g(t) f_T(t) dt \rightarrow$ We'll use the Gamma-function recurrence.

$$\Gamma(n) = (n-1)\Gamma(n-1), \quad \Gamma(n) = (n-1)(n-2)\Gamma(n-2)$$

Take $g(t) = \frac{1}{t} \rightarrow E[\frac{1}{T}] = \int_0^\infty \frac{1}{t} f_T(t) dt$, plug in the Gamma PDF

$$E[\frac{1}{T}] = \int_0^\infty \frac{1}{t} \left(\frac{\theta^n}{\Gamma(n)} t^{n-1} e^{-\theta t} \right) dt = \frac{\theta^n}{\Gamma(n)} \int_0^\infty t^{n-2} e^{-\theta t} dt$$

multiplying by $\frac{1}{t}$ turns t^{n-1} into t^{n-2} , which looks like a Gamma integral

Now let's evaluate the integral by substitution $u = \theta t \rightarrow t = u/\theta$ and $dt = \frac{du}{\theta}$. substitute $\rightarrow$

$$\int_0^\infty t^{n-2} e^{-\theta t} dt = \int_0^\infty \left(\frac{u}{\theta} \right)^{n-2} e^{-u} \left(\frac{du}{\theta} \right) = \frac{1}{\theta^{n-1}} \int_0^\infty u^{n-2} e^{-u} du \rightarrow \int_0^\infty u^{n-2} e^{-u} du = \Gamma(n-1) \rightarrow$$

$$\int_0^\infty t^{n-2} e^{-\theta t} dt = \frac{\Gamma(n-1)}{\theta^{n-1}} \rightarrow \text{plug back} \rightarrow E[\frac{1}{T}] = \frac{\theta^n}{\Gamma(n)} \cdot \frac{\Gamma(n-1)}{\theta^{n-1}} = \theta \cdot \frac{\Gamma(n-1)}{\Gamma(n)}$$

using $\Gamma(n) = (n-1)\Gamma(n-1)$: $\frac{\Gamma(n-1)}{\Gamma(n)} = \frac{1}{n-1}$ Therefore $E[\frac{1}{T}] = \frac{\theta}{n-1}$, $n > 1$

7.2 Compute $E[\frac{1}{T^2}]$. take $g(t) = \frac{1}{t^2} \rightarrow E[\frac{1}{T^2}] = \int_0^\infty \frac{1}{t^2} f_T(t) dt$

$$\rightarrow E\left[\frac{1}{T^2}\right] = \int_0^{\infty} \frac{1}{t^2} \left(\frac{\theta^n}{\Gamma(n)} t^{n-1} e^{-\theta t} \right) dt = \frac{\theta^n}{\Gamma(n)} \int_0^{\infty} t^{n-3} e^{-\theta t} dt$$

$$\rightarrow E\left[\frac{1}{T^2}\right] = \int_0^{\infty} \frac{1}{t^2} \left(\frac{\theta^n}{\Gamma(n)} t^{n-1} e^{-\theta t} \right) dt = \frac{\theta^n}{\Gamma(n)} \int_0^{\infty} t^{n-3} e^{-\theta t} dt \xrightarrow[t = \frac{u}{\theta}]{u = \theta t} \int_0^{\infty} t^{n-3} e^{-\theta t} dt = \rightarrow$$

$$\rightarrow \int_0^{\infty} \left(\frac{u}{\theta} \right)^{n-3} e^{-u} \left(\frac{du}{\theta} \right) = \frac{1}{\theta^{n-2}} \int_0^{\infty} u^{n-3} e^{-u} du = \Gamma(n-2) \rightarrow \int_0^{\infty} t^{n-3} e^{-\theta t} dt = \frac{\Gamma(n-2)}{\theta^{n-2}}$$

$$E\left[\frac{1}{T^2}\right] = \frac{\theta^n}{\Gamma(n)} \cdot \frac{\Gamma(n-2)}{\theta^{n-2}} = \frac{\theta^2 \cdot \Gamma(n-2)}{\Gamma(n)} \xrightarrow[\text{twice}]{\text{recurrence}} \Gamma(n) = (n-1)(n-2)\Gamma(n-2)$$

$$\frac{\Gamma(n-2)}{\Gamma(n)} = \frac{1}{(n-1)(n-2)} \xrightarrow{\text{therefore}} E\left[\frac{1}{T^2}\right] = \frac{\theta^2}{(n-1)(n-2)}, \quad n > 2$$

Now that we have $E\left[\frac{1}{T}\right] = \frac{\theta}{n-1}$ and $E\left[\frac{1}{T^2}\right] = \frac{\theta^2}{(n-1)(n-2)}$

lets now compute $E[\hat{\theta}]$ and $\text{Var}(\hat{\theta})$. Since $\hat{\theta} = \frac{n}{T}$.

$$E[\hat{\theta}] = nE\left[\frac{1}{T}\right] = n \cdot \frac{\theta}{n-1} = \frac{n}{n-1} \theta \quad [\text{mean}]$$

[Variance] $\text{Var}(\hat{\theta}) = E[\hat{\theta}^2] - (E[\hat{\theta}])^2$

$$\begin{aligned} & \begin{array}{c} E\left[\frac{n^2}{T^2}\right] \\ \downarrow \\ n^2 E\left[\frac{1}{T^2}\right] \\ \downarrow \\ n^2 \cdot \frac{\theta^2}{(n-1)(n-2)} \end{array} \quad \begin{array}{c} \swarrow \\ \ominus \\ \left(\frac{n}{n-1} \theta \right)^2 \end{array} \\ \text{Var}(\hat{\theta}) = & n^2 \cdot \frac{\theta^2}{(n-1)(n-2)} - \left(\frac{n}{n-1} \theta \right)^2 = \frac{n^2}{(n-1)^2(n-2)} \theta^2 ; \text{ now lets show that } \text{Var}(\hat{\theta}) \rightarrow 0 \end{aligned}$$

Looking at $\frac{n^2}{(n-1)^2(n-2)} \sim \frac{n^2}{n^2 \cdot n} = \frac{1}{n} \rightarrow 0$ so $\text{Var}(\hat{\theta}) \rightarrow 0$ as $n \rightarrow \infty$

Then
$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}$$

Step ② Sample (empirical) mean and variance. $E_{\hat{f}}[X]$ = sample mean, $\text{Var}_{\hat{f}}[X]$ = sample variance with denominator n .

$$E_{\hat{f}}[X] \equiv \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \text{Var}_{\hat{f}}(X) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \text{Std}_{\hat{f}}(X) = \sqrt{\text{Var}_{\hat{f}}(X)}$$

Step ③ Set up the two MOM Equations. Using Wasserman Def 9.3. "population Moment = Sample Moment" and the Uniform mean/variance formulas. Let $\tilde{a}, \tilde{b}$ be the MoM estimators. Then:

$$E_{\hat{f}}[X] = \frac{\tilde{a} + \tilde{b}}{2}, \quad \text{Var}_{\hat{f}}(X) = \frac{(\tilde{b} - \tilde{a})^2}{12}$$

Step ④ Solve the system.

Introduce the width parameter $\tilde{c} = \tilde{b} - \tilde{a} \rightarrow$ The variance equation becomes

$$\text{Var}_{\hat{f}}(X) = \frac{\tilde{c}^2}{12} \rightarrow \tilde{c}^2 = 12 \text{Var}_{\hat{f}}(X) \rightarrow \sqrt{\quad} \text{ (and choose the positive root because } \tilde{c} = \tilde{b} - \tilde{a} > 0 \text{)}$$

$$\tilde{c} = \sqrt{12} \text{Std}_{\hat{f}}(X) = 2\sqrt{3} \text{Std}_{\hat{f}}(X)$$

Use the mean equation to solve for $\tilde{a}$ and $\tilde{b}$

From the mean equation: $E_{\hat{f}}[X] = \frac{\tilde{a} + \tilde{b}}{2} \rightarrow \tilde{a} + \tilde{b} = 2E_{\hat{f}}[X]$

Now we have the system:

$$\tilde{a} + \tilde{b} = 2E_{\hat{f}}[X], \quad \tilde{b} - \tilde{a} = \tilde{c} = 2\sqrt{3} \text{Std}_{\hat{f}}(X)$$

$$(\tilde{a} + \tilde{b}) + (\tilde{b} - \tilde{a}) = 2\tilde{b} = 2E_{\hat{f}}[X] + 2\sqrt{3} \text{Std}_{\hat{f}}(X)$$

$$\tilde{a} = E_{\hat{f}}[X] - \sqrt{3} \text{Std}_{\hat{f}}(X)$$

Let $\bar{X} = E_{\hat{f}}[X]$ and $S = \text{Std}_{\hat{f}}(X)$. Then the method of moments estimators are

$$\tilde{a} = \bar{X} - \sqrt{3}S, \quad \tilde{b} = \bar{X} + \sqrt{3}S$$

Part (b)

Step ①: Write the uniform (a, b) the density is constant on the interval and 0 outside.

A clean way to write this is: $f(x; a, b) = \frac{1}{b-a} I_{(a,b)}(x)$.

where $I_{(a,b)}(x) = 1$ if $x \in (a, b)$ and 0 otherwise.

Step ② $L(a, b)$ By definition 9.7 for observed data $x_1, \dots, x_n$. $L(a, b) = \prod_{i=1}^n f(x_i; a, b)$

$$L(a, b) = \prod_{i=1}^n \left(\frac{1}{b-a} I_{(a,b)}(x_i) \right)$$

Now let's separate the product:

• The factor $\frac{1}{b-a}$ does not depend on i , so multiplying it n times gives $\left(\frac{1}{b-a}\right)^n$

• The indicators multiply to $\prod_{i=1}^n I_{(a,b)}(x_i)$

so
$$L(a,b) = \left(\frac{1}{b-a} \right)^n \prod_{i=1}^n I_{(a,b)}(x_i)$$

Step ③ Understand what the indicator product means

$$\prod_{i=1}^n I_{(a,b)}(x_i)$$

• If any $x_i \notin (a,b)$ then $I_{(a,b)}(x_i) = 0$, and the whole product becomes 0.

• The product equals 1 only if every $x_i \in (a,b)$

So:

$$\prod_{i=1}^n I_{(a,b)}(x_i) = \begin{cases} 1, & \text{if all } x_i \in (a,b) \\ 0, & \text{otherwise} \end{cases}$$

equivalently "all $x_i \in (a,b)$ " means $a < \min_i x_i$ and $\max_i x_i < b$

Let $x_{(1)} = \min_i x_i$, $x_{(n)} = \max_i x_i$.

Then the Likelihood becomes the piecewise function
$$L(a,b) = \begin{cases} \left(\frac{1}{b-a} \right)^n, & a < x_{(1)} \text{ and } x_{(n)} < b \\ 0, & \text{otherwise} \end{cases}$$

Step ④ Maximize $L(a,b)$ over (a,b)

4.1. We must stay in the region where $L(a,b) \neq 0$

so the MLE must satisfy the constraints $a \leq x_{(1)}$ and $b \geq x_{(n)}$

4.2 Inside that feasible region, maximize $\left(\frac{1}{b-a} \right)^n$

observe:

• $n > 0 \rightarrow$ maximizing $\left(\frac{1}{b-a} \right)^n$ is the same as maximizing $\frac{1}{b-a}$

• maximizing $\frac{1}{b-a}$ is the same as minimizing $b-a$

So to maximize the Likelihood, we want the interval (a,b) to be as short as possible, while still containing all the data.

4.3 The smallest possible interval that contains all data.

This happens when we push: a as large as allowed $\Rightarrow a = x_{(1)}$

b as small as allowed $\Rightarrow b = x_{(n)}$

This yields the minimal width $b-a = x_{(n)} - x_{(1)}$, hence the largest likelihood

Therefore,

$$\hat{a} = x_{(1)} = \min_i x_i, \quad \hat{b} = x_{(n)} = \max_i x_i$$

Part (c)

Step ① Identify the parameter vector and the function g

"original" $\theta = (a, b) \rightarrow$ what we want $\tau = \frac{a+b}{2} \rightarrow g(a, b) = \frac{a+b}{2} \rightarrow \tau = g(\theta)$

Step ② Apply equivariance

$$\hat{\tau} = g(\hat{a}, \hat{b}) = \frac{\hat{a} + \hat{b}}{2} \rightarrow \text{substitute } \hat{a} = X_{(1)} \text{ and } \hat{b} = X_{(n)} \rightarrow \hat{\tau} = \frac{X_{(1)} + X_{(n)}}{2}$$

Part (d)

Compute $MSE(\hat{\tau})$ Analytically.

We have $\hat{\tau} = \bar{X}$

For Uniform (a, b) $Var(X_1) = \frac{(b-a)^2}{12} \rightarrow a=1, b=3 \rightarrow b-a=2, (b-a)^2=4$ hence

$$Var(X_1) = \frac{4}{12} = \frac{1}{3}$$

$$Var(\bar{X}) = Var(X_1) / n ; n=10$$

$$MSE(\hat{\tau}) = \frac{Var(X_1)}{n} = \frac{\frac{1}{3}}{10} = \frac{1}{30} \approx 0.033$$

so for $a=1, b=3, n=10$

Analytic approach $MSE(\hat{\tau}) = \frac{1}{30} \approx 0.033$

Simulation approach $MSE(\hat{\tau}) \approx 0.015$

In this setting, the MLE of the mean has a smaller MSE than the non parametric plug-in estimator.

6a. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta, 1)$ Define,

$$Y_i = \begin{cases} 1 & \text{if } X_i > 0 \\ 0 & \text{if } X_i \leq 0 \end{cases} \quad \text{let } \psi = \mathbb{P}(Y_1 = 1)$$

② Find the MLE for ψ

Step ① Express ψ as a function of θ . Since $X_1 \sim N(\theta, 1)$, standardize:

$$Z = \frac{X_1 - \theta}{1} \sim N(0, 1) \quad \text{now compute } \psi = P(X_1 > 0) = P\left(\frac{X_1 - \theta}{1} > \frac{0 - \theta}{1}\right) = P(Z > -\theta)$$

$$P(Z > -\theta) = \Phi(\theta) \rightarrow \psi = g(\theta) = \Phi(\theta)$$

step ② Find the MLE of θ

$$\textcircled{a} \quad L(\theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_i - \theta)^2}{2}\right).$$

⑥ log Likelihood

$$\ell(\theta) = \log L(\theta) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2$$

⑦ Differentiate and set = 0

$$\text{Diff with respect of } \theta \quad \ell'(\theta) = -\frac{1}{2} \cdot 2 \sum_{i=1}^n (x_i - \theta) (-1) = \sum_{i=1}^n (x_i - \theta)$$

set $\ell'(\theta) = 0$

$$\sum_{i=1}^n (x_i - \theta) = 0 \Rightarrow \sum_{i=1}^n x_i - n\theta = 0 \Rightarrow \hat{\theta} = \bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

⑧ second derivative check (Max)

$$\ell''(\theta) = -n < 0 \text{ so the critical point is a maximum.}$$

step ③ Use equivariance to get the MLE of ψ . We found $\psi = g(\theta) = \Phi(\theta)$

By Wasserman Thm 9.14. the MLE of ψ is $\hat{\psi} = g(\hat{\theta}) = \Phi(\hat{\theta}) = \Phi(\bar{X})$

Then $\hat{\psi}_{MLE} = \Phi(\bar{X})$ where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

Problem 5. Determine the MoM for the Weibull d. whose PDF is: $k > 0$ and $\lambda > 0$ shape $k > 0$ and scale $\lambda > 0$

$$f(x|k, \lambda) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{x}{\lambda}\right)^k\right], & x > 0 \\ 0, & x \leq 0. \end{cases}$$

Step ① Compute the general moment $E[X^r] \rightarrow E[X^r] = \int_0^{\infty} x^r f(x|k, \lambda) dx$

plug in the Weibull PDF: $E[X^r] = \int_0^{\infty} x^r \left[\frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} \right] dx$

$$\frac{k}{\lambda} \cdot \frac{x^{k-1}}{\lambda^{k-1}} = \frac{k}{\lambda^k} x^{k-1}$$

$$E[X^r] = \int_0^{\infty} \frac{k}{\lambda^k} x^{r+k-1} e^{-(x/\lambda)^k} dx$$

let $u = \left(\frac{x}{\lambda}\right)^k \rightarrow x = \lambda u^{1/k}, dx = \lambda \cdot \frac{1}{k} u^{1/k-1} du$

Also, $x^{r+k-1} = (\lambda u^{1/k})^{r+k-1} = \lambda^{r+k-1} u^{(r+k-1)/k} \rightarrow \text{substitute} \rightarrow$

$$E[X^r] = \int_0^{\infty} \frac{k}{\lambda^k} \left(\lambda^{r+k-1} u^{(r+k-1)/k} \right) e^{-u} \left(\lambda \frac{1}{k} u^{1/k-1} \right) du$$

canceling constants and combining powers of u

constants $\frac{k}{\lambda^k} \cdot \lambda^{r+k-1} \cdot \lambda \cdot \frac{1}{k} = \lambda^r$ Powers of u $u^{(r+k-1)/k} \cdot u^{1/k-1} = u^{\frac{r+k-1}{k} - 1} = u^{\frac{r+k-1}{k} - \frac{k}{k}} = u^{\frac{r}{k}}$

$$E[X^r] = \lambda^r \int_0^{\infty} u^{r/k} e^{-u} du \rightarrow \text{compare with the Gamma definition } \Gamma(k) = \int_0^{\infty} u^{k-1} e^{-u} du$$

here $u^{r/k} = u^{(1+r/k)-1}$, so $\alpha = 1 + r/k$ therefore $\int_0^{\infty} u^{r/k} e^{-u} du = \Gamma\left(1 + \frac{r}{k}\right) \rightarrow$

$$E[X^r] = \lambda^r \Gamma\left(1 + \frac{r}{k}\right)$$

Step ② MoM. we need 2 equations because we have 2 unknowns (k, λ)

for $r=1$ $E[X] = \lambda \Gamma\left(1 + \frac{1}{k}\right)$

for $r=2$ $E[X^2] = \lambda^2 \Gamma\left(1 + \frac{2}{k}\right)$

$$\hat{\alpha}_1 = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}$$

$$\hat{\alpha}_2 = \frac{1}{n} \sum_{i=1}^n x_i^2 = \bar{X}^2$$

Set population = sample $\rightarrow$

$$\bar{X} = \lambda \Gamma\left(1 + \frac{1}{k}\right),$$

$$\bar{X}^2 = \lambda^2 \Gamma\left(1 + \frac{2}{k}\right)$$

Step ③ Solve the system for (κ, λ)

$$\bar{X} = \lambda \Gamma\left(1 + \frac{1}{\kappa}\right)$$

divide both sides by $\Gamma\left(1 + \frac{1}{\kappa}\right)$:

$$\lambda = \frac{\bar{X}}{\Gamma\left(1 + \frac{1}{\kappa}\right)}$$

$$\bar{X}^2 = \lambda^2 \Gamma\left(1 + \frac{2}{\kappa}\right)$$

→ substitute $\lambda = \frac{\bar{X}}{\Gamma\left(1 + \frac{1}{\kappa}\right)}$

$$\bar{X}^2 = \left(\frac{\bar{X}}{\Gamma\left(1 + \frac{1}{\kappa}\right)}\right)^2 \Gamma\left(1 + \frac{2}{\kappa}\right)$$

rearrange by dividing both sides by $\bar{X}^2$

$$\frac{\bar{X}^2}{\bar{X}^2} = \frac{\Gamma\left(1 + \frac{2}{\kappa}\right)}{\left[\Gamma\left(1 + \frac{1}{\kappa}\right)\right]^2}$$

$$\text{let } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\bar{X}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

$\tilde{\kappa}$ is the solution to

$$\frac{\bar{X}^2}{\bar{X}^2} = \frac{\Gamma\left(1 + \frac{2}{\tilde{\kappa}}\right)}{\left[\Gamma\left(1 + \frac{1}{\tilde{\kappa}}\right)\right]^2}$$

$$\text{Then } \tilde{\lambda} = \frac{\bar{X}}{\Gamma\left(1 + \frac{1}{\tilde{\kappa}}\right)}$$

Problem 6. MLE and Empirical MSE for the Normal distribution.

Let $X_1, \dots, X_n$ be iid from a Normal Distribution with unknown mean μ and known variance σ^2

1. Write the likelihood function $L(\mu)$ and log-likelihood $\ell(\mu)$

2. Show that maximizing $\ell(\mu)$ is equivalent to minimizing the empirical mean squared error $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$

3. Use the result to find the MLE $\hat{\mu}$.

① write the normal PDF (with known σ^2). For one observation X_i , $f(X_i | \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right)$

② Multiply over $i=1, \dots, n$ (likelihood definition)

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right) \rightarrow L(\mu) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2\right)$$

③ Take logs to get $\ell(\mu)$

$$\ell(\mu) = \log L(\mu) = n \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \quad \text{since } \log(1/\sqrt{2\pi\sigma^2}) = -\frac{1}{2} \log(2\pi\sigma^2)$$

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

② Show maximizing $l(\mu) \Leftrightarrow$ minimizing empirical MSE $\frac{1}{2} \sum (X_i - \mu)^2$

$$l(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

Identify what depends on μ

- $-\frac{n}{2} \log(2\pi\sigma^2)$ does not involve μ , then it is a constant

- The only μ -dependent part is $-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$

Then the maximizer of $l(\mu)$ is the same as the maximizer of $-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$

Now let's turn maximize a negative quadratic into minimize the quadratic.

Because $\sigma^2 > 0$, the constant $\frac{1}{2\sigma^2} > 0$. So maximizing $-\frac{1}{2\sigma^2} \sum (X_i - \mu)^2$ is the same as

$$\text{minimizing } \sum_{i=1}^n (X_i - \mu)^2$$

The problem's "empirical MSE" is $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ since $\frac{1}{n} > 0$, minimizing $\sum (X_i - \mu)^2$ is equivalent to minimizing $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$. Therefore $\arg \max_{\mu} l(\mu) = \arg \min_{\mu} \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$

③ We now minimize $Q(\mu) = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ because $\frac{1}{n}$ is a positive constant, it has the same minimizer as $S(\mu) = \sum_{i=1}^n (X_i - \mu)^2$

④ Differentiate $S(\mu)$

$$\text{Expand each term } (X_i - \mu)^2 \xrightarrow{\text{Derivative w.r.t } \mu} \frac{d}{d\mu} (X_i - \mu)^2 = 2(X_i - \mu) \cdot (-1) = -2(X_i - \mu)$$

⑤ Set derivative = 0 and solve. Set $S'(\mu) = 0$

$$-2 \sum_{i=1}^n (X_i - \mu) = 0 \rightarrow \sum_{i=1}^n (X_i - \mu) = 0 \rightarrow \sum X_i - n\mu = 0 \rightarrow \hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n X_i$$

⑥ Confirm it's a minimum (and hence a maximum of l)

$$S'(\mu) = -2 \sum (X_i - \mu) = -2(\sum X_i - n\mu) \rightarrow S''(\mu) = 2n > 0$$

So $S(\mu)$ is convex and the critical point is the unique minimum. Therefore it also maximizes $l(\mu)$